UPI Fraud Detection: Project Summary

TO: Fraud Risk Team Lead
SUBJECT: Findings from Synthetic UPI Transaction Analysis

1. The Business Question

Which transaction patterns in a UPI payment flow are most predictive of fraud, and what is the precision-recall trade-off at different risk thresholds?

2. Dataset Context & Limitations

The analysis was conducted on a 100,000-row synthetic UPI dataset. The fraud rate is 23.99%. A critical limitation was identified: the dataset includes pre-computed historical features (e.g., Transaction_Frequency) that may contain future information leakage. Furthermore, we removed Transaction_Status because it is a post-transaction metric unavailable at real-time scoring.

3. Analytical Investigations

1. SQL Layer: We utilized window functions to analyze transaction velocity. Results showed a flat fraud rate across all time-gap buckets, indicating no velocity signal.
2. Machine Learning: We trained Logistic Regression and Random Forest models using StratifiedGroupKFold by Customer_ID. This prevents the model from memorizing specific customers.

4. Findings & Null Results

Both models achieved an Out-of-Fold ROC-AUC of ~0.50 (equivalent to random chance). The baseline Dummy Classifier scored 0.4986, proving the models learned nothing.

Business Cost Simulation (using OOF predictions):
- At Threshold 0.3: The model flags all 100,000 transactions. Investigation cost = Rs 2,000,000.
- At Threshold 0.7: The model misses all 23,986 frauds. Missed fraud cost = Rs 11,993,000.
The model fails to provide a cost-effective fraud filter.

5. Business Implications

Because the model has no predictive power, deploying it would result in massive false-positive investigation costs or huge missed-fraud losses. The data science pipeline is robust and leakage-free, but the synthetic dataset lacks the complex network and behavioral signals of real UPI traffic.

Code and reproducibility instructions: github.com/bosesrinjoy24/upi-fraud-risk-project